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Hospital Appointment and Queue System

↳ Purpose:-

The purpose of the hospital Appointment and Queue System is to make hospital visits more organized, faster and fairer.

The system should

- ⇒ Allow Patients to book appointment without visiting or calling hospital.
- ⇒ Reduce long waiting line at reception desk.
- ⇒ Help doctors see their daily schedule and Patient queue.
- ⇒ Notify Patient about appointment

date changes, cancellation and delay.
⇒ Help Staff prioritize emergency cases
⇒ Reduce missed appointment by sending a reminder.

↳ Users:-

⇒ **Patients:-** Registered, Book, cancell and reschedule appointment, check in, receive queue number and view estimating waiting time.

⇒ **Receptionists:-** Registered, walk-in patients, verify appointments, check patients in and manage a queue.

⇒ **Doctors:-** view daily appointment, see current Patient queue, mark consultant as complete and report delay or unavailability.

⇒ **Nurses:-** Check patient arrival status, upgrade basic triage information and help prioritize emergency case.

⇒ **Hospital administrator:-** Manage department, doctors, schedules, user account report and system rule.

⇒ **IT support staff:-** Maintain the system, solve technical problem protect data and make back up.

L> Stakeholders:-

Stakeholders are individual or groups affected by the system

⇒ **Patients and families:** They want short waiting times, correct appointment, privacy and clear information.

⇒ **Doctors:-** They need accurate patient list and a schedule that don't create unrealistic workload.

⇒ **Nurses and receptionists:-** They need an easy system that reduce manual work and prevent booking errors.

⇒ **Hospital management:-** They need better use of doctors, room, staff and hospital resource.

⇒ **IT department:-** They are responsible for maintenance, security, backup.

⇒ **Insurance provider:-** They may need accurate appointment, patient and billing related record.

⇒ **Software developer:-** They design, update and support the system.

L> Functional Requirements:-

Functional

requirement describe the action of system

- ⇒ Patients can registered and login by using mobile number and Email.
- ⇒ Patients can book an appointment by choosing department, doctor, and time.
- ⇒ Patients can cancel or reschedule the appointment.
- ⇒ System create a queue and show the current queue number and estimated waiting time.
- ⇒ System give you a notification in any case delay or arrival time.

↳ Non functional requirement:-

Non functional requirement describe quality and performance.

- ⇒ The system must protect patient account detail and health related information.
- ⇒ The system should be available all time.
- ⇒ The system should response quickly.
- ⇒ The interface must be simple for different ages people.
- ⇒ The system must support a growing in many branches without becoming slow.

↳ Risk:

⇒ Data breach:- Unauthorized person could access patient personal detail and health information.

⇒ system failure:- Patient unable to book appointment or check in. Staff may lose access to live queue during busy period.

⇒ Incorrect queue priority or booking Error:-

A patient could be given the wrong information or urgent patient may be delay.

↳ Quality attributes:-

⇒ Reliability:-

The system should be correctly save information or detail. It must not double booking.

⇒ Security and Privacy:-

System handle sensitive health related data and able to access the information.

⇒ useability:-

Patient must complete task for booking appointment.

L> where AI could be used:-

- ⇒ waiting time prediction
- ⇒ No show prediction
- ⇒ Smart appointment scheduling
- ⇒ Chatbot assistant
- ⇒ Demand forecasting

L> what could go wrong if ai is used incorrectly:-

- ⇒ Incorrect waiting time prediction
- ⇒ Unfair priority decision
- ⇒ Privacy and data misuse
- ⇒ Unsafe chatbot advice
- ⇒ Too much dependance on automation